

INFO 5602
Information Visualization

“Think Like a Designer”

Agenda:

- “Form follows function”
- Traditional design concepts
 - Affordances
 - Accessibility
 - Aesthetics
- Gaining audience acceptance of your visualization product
- Group exercise/learning:
 - Continue 50-chart visualization challenge discussion

“Form follows function”

- Every chart, dashboard, visualization etc is a **product that you design for**
- In **product design**, a key concept is that “form follows function”
 - Think about what we want our audiences to be able to *do* with the data (function) and then create a visualization (form) that will allow for this with the *most ease* (*i.e. lowest cognitive load*)
- Particular design concepts apply to the production of your visualization product, including:
 - Affordances
 - Accessibility
 - Aesthetics
- Beyond these design concepts, we need a plan to **gain audience acceptance for our visualization products**

Affordances

Traditional design concepts: **Affordances**

- Affordances are aspects inherent to the design of a (visual) product that make it obvious how the product is to be used
- Affordances suggest how a product is to be interacted with or operated
- When **sufficient affordances are present**, good design fades into the background, and you do not even notice it
- Options to **afford** understanding of your visual product by stakeholders:
 - Highlight the important stuff
 - Eliminating distractions
 - Creating a visual hierarchy of information

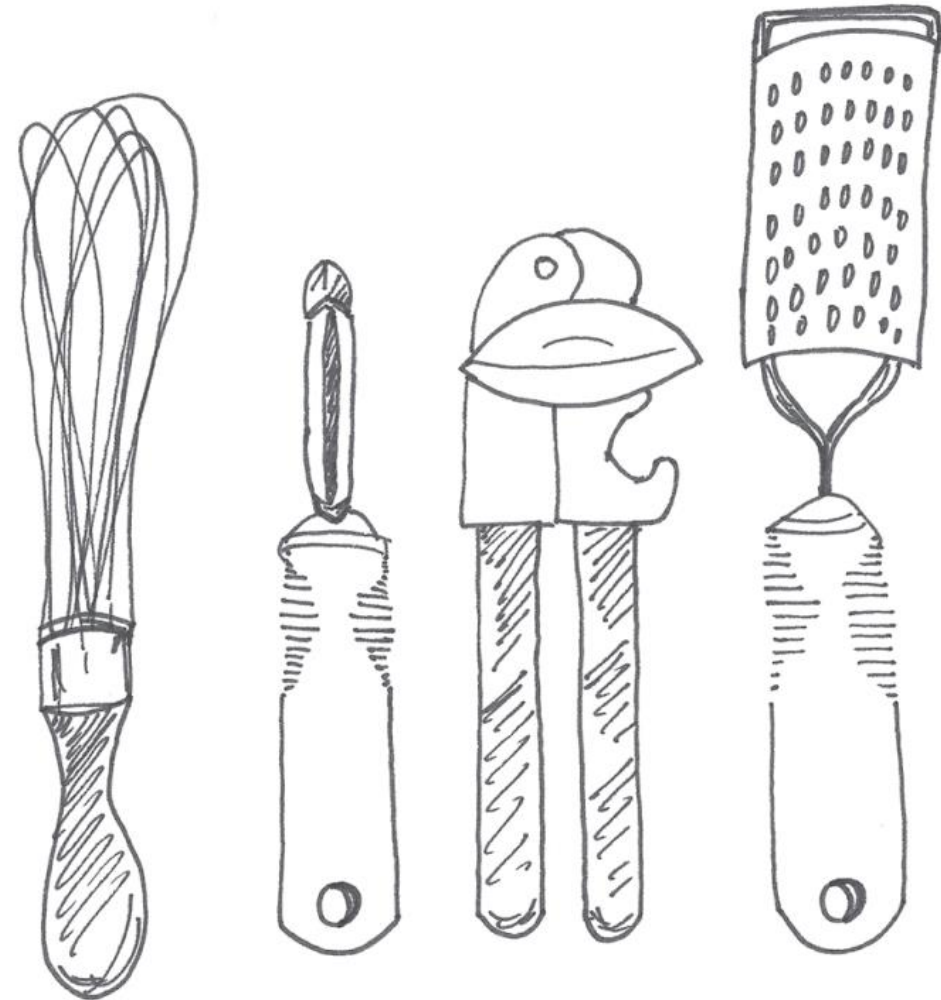


FIGURE 5.1 OXO kitchen gadgets

Highlight the important stuff to *afford* understanding of your visualization product

- Lidwell et al (2003) recommends that at **most 10% of your visualizations should be highlighted**
- Use **bold**, *italics*, underlining, CASE, **Color**, Size etc. to highlight important stuff on your visual product to your stakeholders
 - **Bold**, *italics*, and underlining: use for titles, labels, captions to differentiate elements. Bolding is generally preferred to italics because it tends to add minimal noise to the design while clearly highlighting chosen elements
 - CASE: use for titles, labels and keywords. Avoid using different fonts as a highlighting technique
 - **Color**: is an effective highlighting technique when used sparingly and generally in concert with other highlighting technique (e.g. **bold**)
 - Size (of a chart element) is another way to attract attention and signal importance

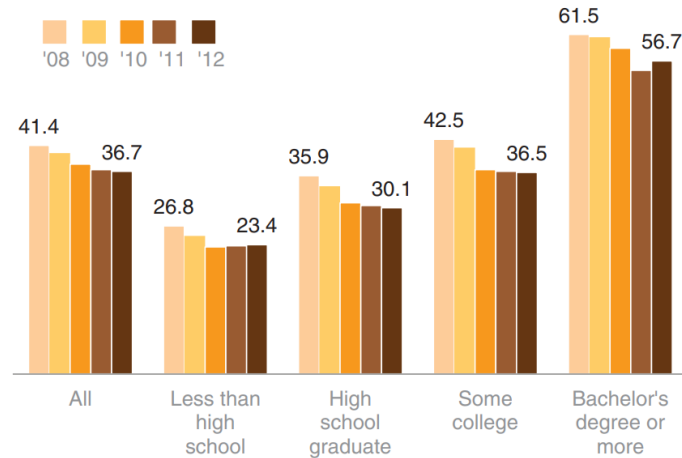
Highlight the important stuff to *afford* understanding of your visualization product

- Use **bold**, *italics*, underlining, CASE, Color, Size etc. to highlight important stuff on your visual product to your stakeholders

“Before”

New Marriage Rate by Education

Number of newly married adults per 1,000 marriage eligible adults



Note: Marriage eligible includes the newly married plus those widowed, divorced, or never married at interview.

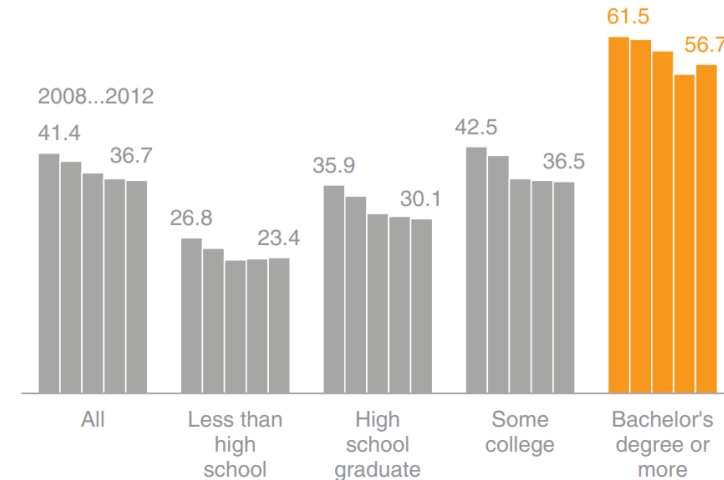
Source: U.S. Census

Adapted from PEW RESEARCH CENTER

“After”

New Marriage Rate by Education

Number of newly married adults per 1,000 marriage eligible adults



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Source: U.S. Census

Adapted from PEW RESEARCH CENTER

Eliminate distractions to *afford* understanding of your visualization product to your stakeholders

- While we highlight the important stuff, we also want to **eliminate distractions**
- “You know you’ve achieved perfection, not when you have nothing more to add, but when you have nothing to take away” Saint-Exupery (1943)
- When it comes to the perfection of design with data visualization, the decision of what to cut or de-emphasize can be more important than what to include or highlight
- To identify distractions, think about both **clutter (noise)** and **context**

Eliminate distractions to *afford* understanding of your visualization product to your stakeholders

- To identify distractions, think about **context, signal and clutter (noise)**
- **Context is what you use to turn exploratory analysis to explanatory analysis i.e. Why? Who? What? How?**
- **Signal is the key message your stakeholders are holding out for**
- **Clutter is what adds cognitive load or noise to your visuals**
- Identify what information is critical – and what is not
- Determine whether there are elements of your chart that might be irrelevant or distracting from your main message – these elements are candidates for elimination

Eliminate distractions to *afford* understanding of your visualization product to your stakeholders

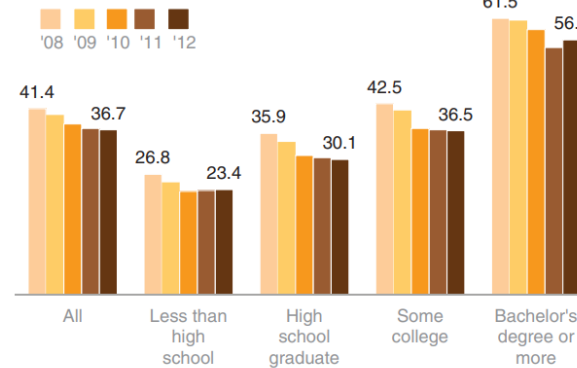
- To identify (potential) distractions think about the following:
 - **Not all data are equally important:** get rid of noncritical data or components
 - **When detail is not needed, summarize:** You should be familiar with the detail, but that does not mean your stakeholders need to be
 - **Ask yourself: would eliminating this chart element change anything?** If No, remove it. Do not be emotionally connected with your charts even if you worked hard to create them
 - **Push necessary, but non-message-impacting items to the background:** Use your knowledge of pre-attentive attributes to de-emphasize

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Source: U.S. Census

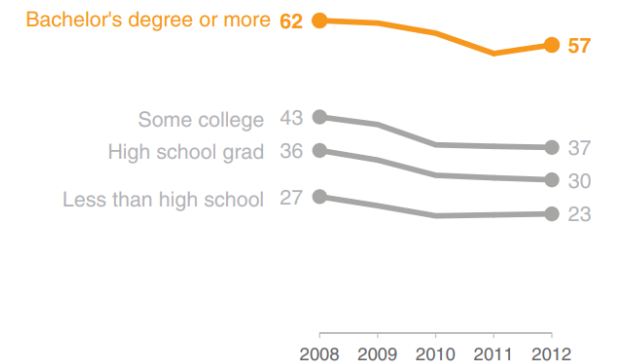
Adapted from PEW RESEARCH CENTER

FIGURE 5.5 Before-and-after

Knafllic (2017)

New marriage rate by education

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Adapted from PEW RESEARCH CENTER

Establish a visual hierarchy of information to *afford* understanding of your visualization product to your stakeholders

- Pre-attentive attributes can be used to create a hierarchy of information to boost the signal in your presentation
- A hierarchy of information is a way of pulling some chart elements to the fore and pushing other elements to the background – **to indicate to our stakeholders the general order in which they should process the information we are communicating** (a.k.a affordances)

Issues vs. Satisfaction by Model

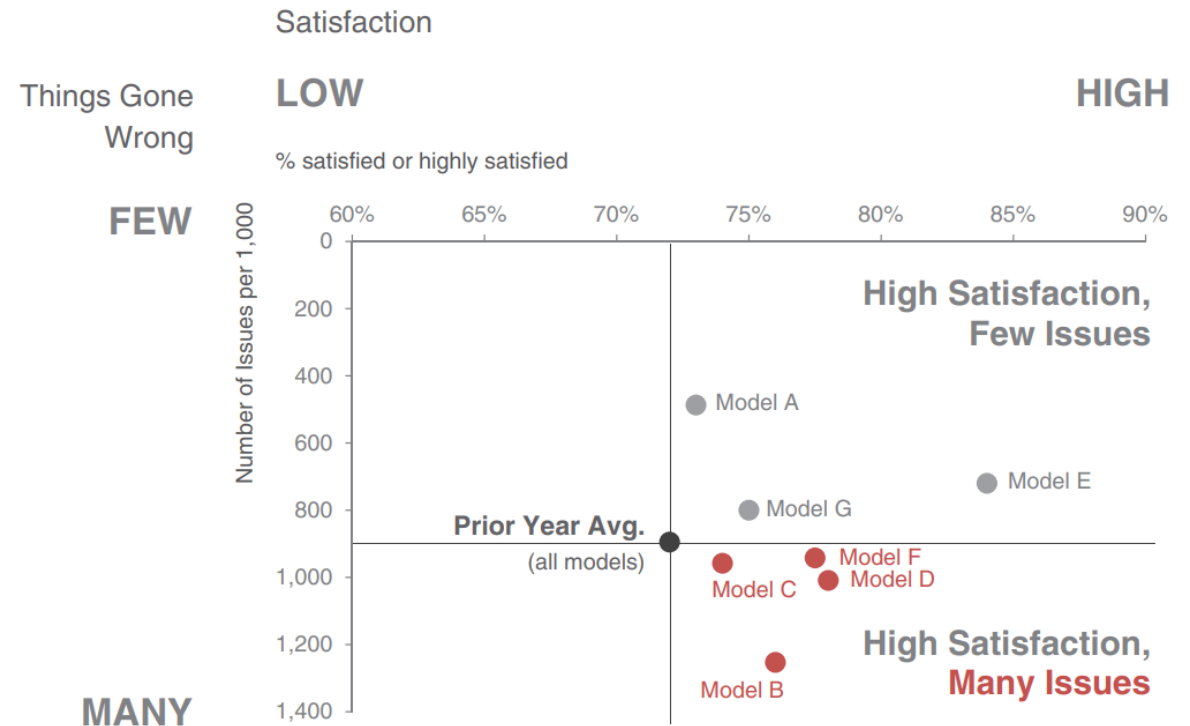


FIGURE 5.6 Clear visual hierarchy of information

Knafllic (2017)

Accessibility

Traditional design concepts: **Accessibility**

- The concept of accessibility says that **product designs should be usable by people of diverse abilities**
- Information visualization **designs should be usable by people of widely varying technical skills**. For example, you might be an engineer, but it should not take someone with an engineering degree to understand your graph (given that your stakeholders might have diverse backgrounds)
- As the expert, it is your responsibility to make your dashboards or charts accessible. **Always assume that your audience has people from heterogeneous backgrounds and skill levels**
- Well-designed information visualization – like any well-designed product – is **easy to interpret and understand**

Traditional design concepts: **Accessibility**

- Well-designed information visualization – like any well-designed product – is easy to interpret and understand
- To improve the **accessibility of your charts**, try the following:
 - Do not overcomplicate your charts or dashboards
 - Use annotations (text) to help your stakeholders quickly understand your message

Do not overcomplicate your charts

- “If it’s hard to read, its hard to do” – Song and Schwarz (2008)
- The more complicated anything looks, the more time your audience **perceives it will take to understand and the less likely they are to spend time to understand it**
- **Make chart text legible:** use a consistent, easy-to-read font
- **Keep it clean:** make your information visualization approachable by using visual affordances
- **Use straightforward language:** choose simple language over complex; choose fewer words over more words; define any specialized language you use to avoid making your audience feel dumb
- **Remove unnecessary complexity:** when making a choice between simple and complicated, favor simple

Use annotations (text) to help your stakeholders quickly understand your message

- Thoughtful **use of text** helps ensure information visualization is accessible
- Text plays several roles when you communicate with data: use it to label, introduce, explain, reinforce, highlight, recommend, **and tell a story**
- Assume every chart needs a title and every axis needs a title
- **Never assume that two different people looking at the same information visualization will draw the same conclusion**
- If there is a conclusion you want your audience to reach, **state it in words**
 - Use pre-attentive attributes to make those words stand out

Use annotations (text) to help your stakeholders quickly understand your message

- Thoughtful **use of text** helps ensure information visualization is **accessible**
- Text plays several roles when you communicate with data: use it to label, introduce, explain, reinforce, highlight, recommend, and **tell a story**
- Annotate important points directly on your charts when that helps affordances, accessibility and reduces cognitive load overall

Peak Break-up Times
According to Facebook status updates

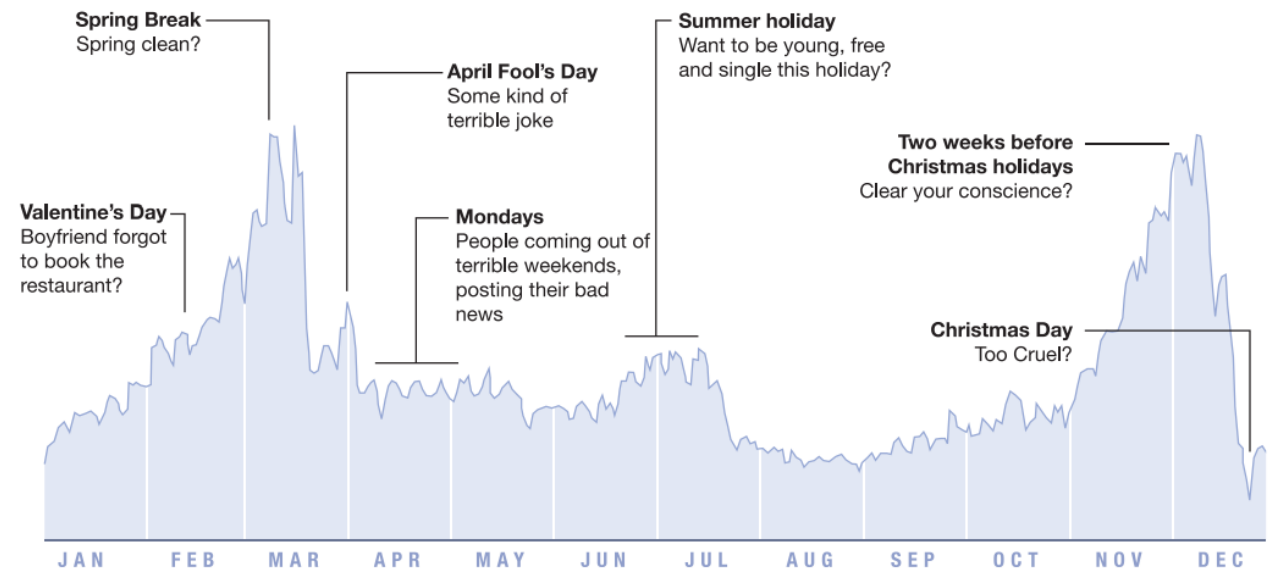


FIGURE 5.7 Words used wisely

Knafllic (2017)

Aesthetics

Traditional design concepts: **Aesthetics**

- People perceive more aesthetic designs as **easier to use** than less aesthetics designs
- According to Knaflitz (2017) studies have shown that more aesthetic designs are:
 - **perceived as easier to use**
 - **more readily accepted and used over time**
 - promote creative thinking and problem solving, and foster positive relationships, making people more tolerant of problems with designs
- Find and use examples of well-designed, good-looking charts and dashboards as templates to create your own.
 - You do not have to re-invent the wheel

Traditional design concepts: **Aesthetics**

- To improve the aesthetics of your chart designs, try the following:
 - **Be smart with color:** use color sparingly and strategically to highlight the important parts of your visuals
 - **Pay attention to alignment:** organize elements on the page to create clean vertical and horizontal lines to establish a sense of unity and cohesion
 - **Leverage white space:** preserve margins; do not stretch your charts to fill the space on your page, or add things simply because you have extra space

Traditional design concepts: **Aesthetics**

- **Be smart with color:** use color sparingly and strategically to highlight the important parts of your visuals

“Before”

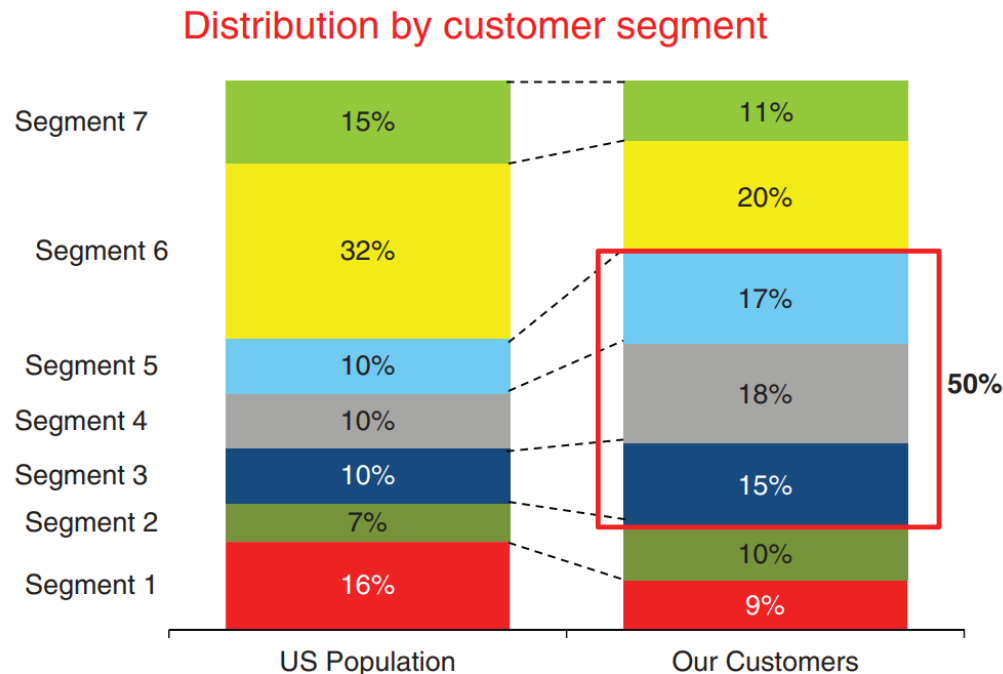


FIGURE 5.12 Unaesthetic design

“After”

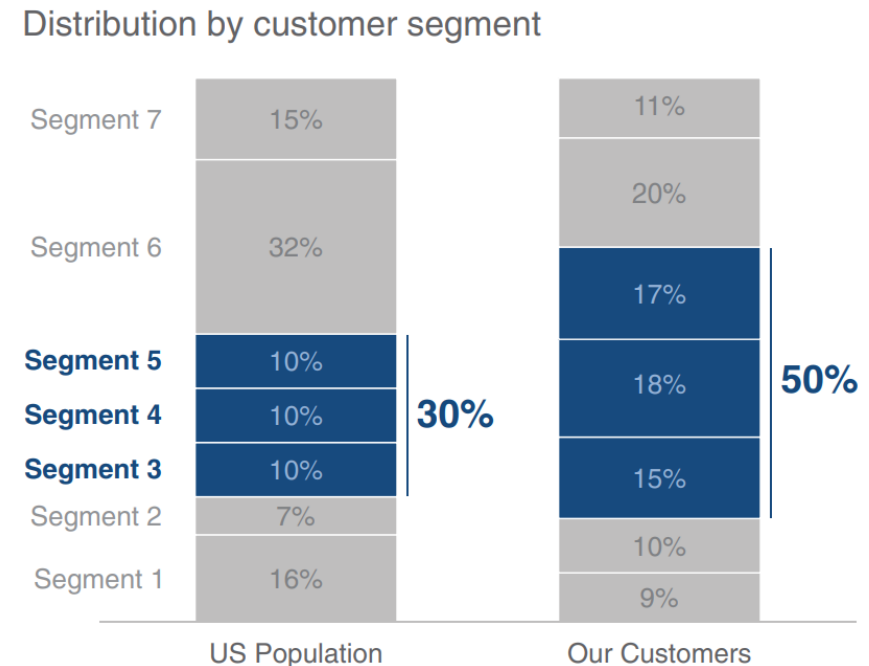


FIGURE 5.13 Aesthetic design

Gaining audience acceptance of
your visualization product

Gaining audience **acceptance** of your visualization product

- For a design to be effective, it must be **accepted** by its intended audience
- Most people **experience some level of discomfort with change partly because of their familiarity with the old** Lidwell et al (2010)
- Many stakeholders tend to hold on to “the way we’ve always done it”
- Tips for gaining acceptance in the design of your data visualization:
 - Articulate the benefits of your new proposed approach to stakeholders
 - Show the side-by-side i.e. “before” versus “after”
 - Provide multiple options and seek input (i.e. meet with your stakeholders *ahead* of presentations to a larger audience)
 - Get **vocal members** of your audience on board *ahead* of presentations to a larger audience

End of presentation